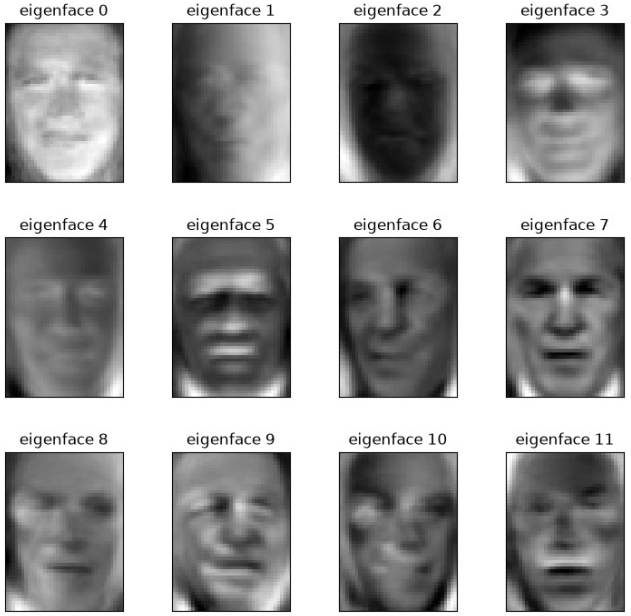
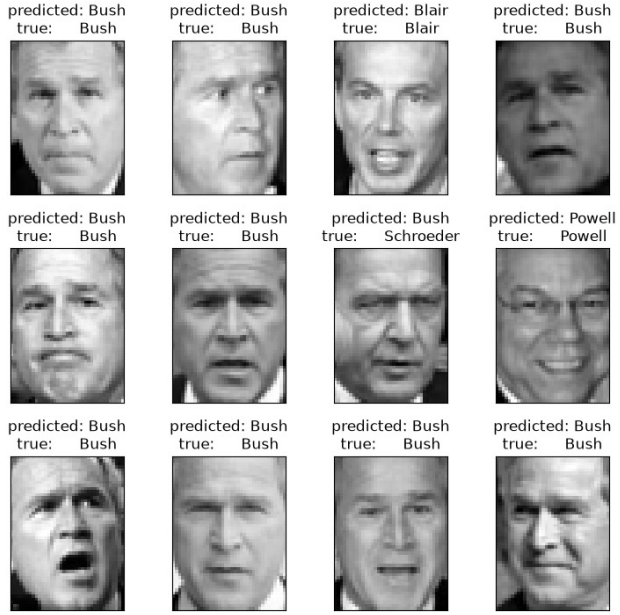
Instructions for Set 6

1. Face detection
2. Eigenface builder
3. Face recognition







Set 6.1 Face detection (not Haar)



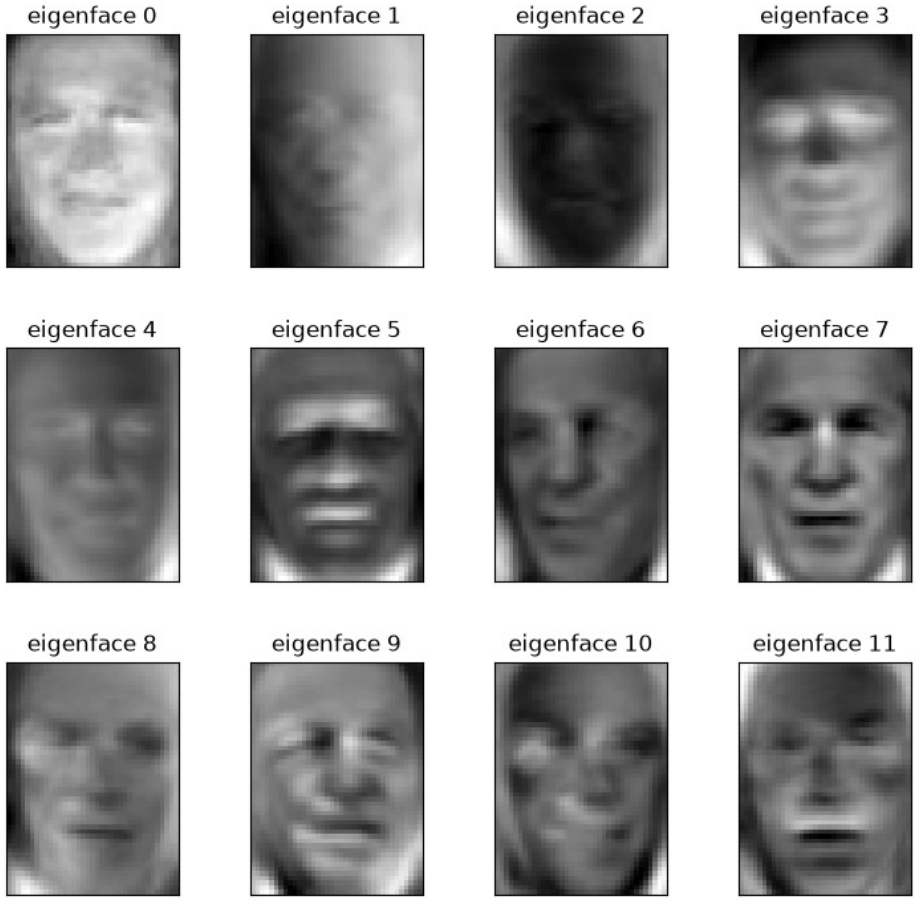
Can you try run it on a webcam stream?

Face detection using Haar and AdaBoost as well as YuNet is in the tutorial notebook

Set 6.2 Eigenfaces

Open folder with images:

1. Save the images to a folder (train and test)
    1. from sklearn.datasets import fetch_lfw_people
    2. don't forget the labels and correspondence (image <-> name pairs)
2. Save the model (or compute every run):
    1. mean, eigenfaces, PCA coefficients (e.g., to a .npy): $\mathbf{w}_i = U_K^\top(\mathbf{x}_i - \boldsymbol{\mu})$
3. Visualize first K eigenfaces



Set 6.3 Face recognition

Open one image from folder (saved in test):

1. Load the saved model (or compute every run):
    1. mean, eigenfaces, PCA coefficients (e.g., to a .npy)
2. Calculate the coefficients for the test image

$$\mathbf{w}_{test} = U_K^\top (\mathbf{x}_{test} - \boldsymbol{\mu})$$

3. Display the closest 2-3 images from train set with names

$$d_i = \|\mathbf{w}_{test} - \mathbf{w}_i\|_2$$

$$i^* = \arg\min_i d_i$$

